

CS 374: Section A - Homework 4, Problem 2

Anirudh Konidala

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Throughout, $\Sigma = \{0, 1\}$, and $\#(a, x)$ denotes the number of occurrences of the symbol a in the string x .

Part (a)

All strings in which every prefix has at least as many 1s as 0s, that is,

$$\{w \in \Sigma^* \mid \#(1, x) \geq \#(0, x) \text{ for every prefix } x \text{ of } w\}.$$

$S \rightarrow \varepsilon \mid 1S \mid 1B0S$	every prefix has at least as many 1s as 0s
$B \rightarrow \varepsilon \mid 1B0B$	balanced, every prefix has at least as many 1s as 0s

For a string u write $d(u) = \#(1, u) - \#(0, u)$. The two non-terminals generate the following languages.

- B generates every string u with $d(u) = 0$ such that $d(x) \geq 0$ for every prefix x of u . These are the strings with equally many 0s and 1s in which no prefix has more 0s than 1s.
- S generates every string w such that $d(x) \geq 0$ for every prefix x of w , which is exactly the language asked for.

Why this is correct. Reading a 1 raises d by one and reading a 0 lowers it by one, so d of a prefix is the running count that the language constrains. Each description is checked in both directions.

Every string B generates is balanced with non-negative prefix counts. This is clear for ε . If $u, v \in B$, then the prefixes of $1u0v$ are the strings $1u'$, with $d = 1 + d(u') \geq 1$, and the strings $1u0v'$, with $d = 1 + 0 - 1 + d(v') \geq 0$. The whole string has $d(1u0v) = 0$.

Every such string is generated by B . Let u be non-empty with $d(u) = 0$ and $d(x) \geq 0$ for every prefix x . Its first symbol is 1, since the prefix 0 would have $d = -1$. Let i be the smallest index with $d(u[1..i]) = 0$; this index exists because $d(u) = 0$, and $i \geq 2$. Then $u[i] = 0$, because $d(u[1..i-1]) \geq 1$ by the choice of i . So

$$u = 1y0z, \quad \text{where } y = u[2..i-1] \text{ and } z = u[i+1..].$$

Here $d(y) = 0$, and every prefix y' of y has $d(y') = d(1y') - 1 \geq 0$, again by the choice of i , so $y \in B$. And z is the rest of u after a prefix with $d = 0$, so $d(z) = 0$ and every prefix of z has $d \geq 0$, so $z \in B$. This is the rule $B \rightarrow 1B0B$, and induction on length finishes the argument.

Every string S generates has non-negative prefix counts. This is clear for ε . If $w \in S$, then every prefix of $1w$ has $d = 1 + d(w') \geq 1$. If $u \in B$ and $w \in S$, then the prefixes of $1u0w$ have $d \geq 1$, then $d = 0$, then $d = d(w') \geq 0$.

Every such string is generated by S . Let w be non-empty with $d(x) \geq 0$ for every prefix x , and apply the hint.

- If no non-empty prefix x of w has $d(x) = 0$, then every non-empty prefix has $d \geq 1$. In particular $w = 1v$, and every prefix v' of v has $d(v') = d(1v') - 1 \geq 0$, so $v \in S$ and w comes from $S \rightarrow 1S$.
- Otherwise let x be the shortest non-empty prefix with $d(x) = 0$. Exactly as in the argument for B , we have $x = 1y0$ with $y \in B$, and the remaining suffix w' has $d(w'') = d(xw'') \geq 0$ for each of its prefixes w'' , so $w' \in S$ and w comes from $S \rightarrow 1B0S$.

Induction on length finishes the argument.

Sources and collaborators: None.

LLM use: I used Claude (Anthropic) to draft these grammars and explain their structure to me. I checked every grammar by hand against the examples in the handout and against several strings inside and outside each language, and typeset the submission myself.

Part (b)

Even-length strings whose second half contains an even number of 1s, that is,

$$\{xy \in \Sigma^* \mid |x| = |y| \text{ and } \#(1, y) \text{ is even}\}.$$

$X \rightarrow 0 \mid 1$	the two one-symbol strings
$E \rightarrow \varepsilon \mid XE0 \mid XO1$	even-length, second half has an even number of 1s
$O \rightarrow XO0 \mid XE1$	even-length, second half has an odd number of 1s

The start symbol is E . The three non-terminals generate the following languages.

- X generates the two strings 0 and 1.
- E generates every string xy with $|x| = |y|$ and $\#(1, y)$ even.
- O generates every string xy with $|x| = |y|$ and $\#(1, y)$ odd.

Why this is correct. Every rule other than $E \rightarrow \varepsilon$ adds one symbol at the far left and one symbol at the far right, so a derivation of a string of length $2n$ applies exactly n of them. The n symbols added on the left, in the order they were added, spell out the first half x , and the n symbols added on the right spell out the second half y from its last symbol backwards. The halves therefore always have equal length, and the only thing the rules need to track is the parity of $\#(1, y)$.

- A symbol 0 placed on the right keeps that parity, which is why $E \rightarrow XE0$ and $O \rightarrow XO0$ leave the non-terminal unchanged.
- A symbol 1 placed on the right flips it, which is why $E \rightarrow XO1$ and $O \rightarrow XE1$ switch between the two non-terminals.
- The base case $E \rightarrow \varepsilon$ is correct because the empty second half contains zero 1s, and O has no ε -rule for the same reason.

Formally, both inclusions follow by induction on n . Each rule preserves the claimed description, as the previous paragraph shows. For the other direction, let xy have $|x| = |y| = n$.

- If $n = 0$, then $\#(1, y) = 0$ is even and $E \rightarrow \varepsilon$ applies.
- If $n \geq 1$, write $x = ax'$ and $y = y'b$ with $a, b \in \Sigma$. Then $|x'| = |y'| = n - 1$, and $\#(1, y) = \#(1, y') + 1$ when $b = 1$, while $\#(1, y) = \#(1, y')$ when $b = 0$. So if $\#(1, y)$ is even, then $x'y'$ is generated by E when $b = 0$ and by O when $b = 1$, and the rules $E \rightarrow XE0$ and $E \rightarrow XO1$ produce xy . The odd case is symmetric.

Sources and collaborators: None.

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Part (c)

$$\{0^a 1^b 0^c 1^d \mid a, b, c, d \geq 1 \text{ and if } a = d \text{ then } b = c\}$$

$S \rightarrow OMI \mid G \mid H$	the language
$G \rightarrow 0G1 \mid 0OIO1$	$0^a 1^b 0^c 1^d$ with $a > d$
$H \rightarrow 0H1 \mid 0IOI1$	$0^a 1^b 0^c 1^d$ with $a < d$
$M \rightarrow 10 \mid 1M0$	$1^n 0^n$ with $n \geq 1$
$O \rightarrow 0 \mid 0O$	0^n with $n \geq 1$
$I \rightarrow 1 \mid 1I$	1^n with $n \geq 1$

The six non-terminals generate the following languages.

- O generates every non-empty string of 0s, that is, every 0^n with $n \geq 1$.
- I generates every non-empty string of 1s, that is, every 1^n with $n \geq 1$.
- M generates every string $1^n 0^n$ with $n \geq 1$.
- G generates every string $0^a 1^b 0^c 1^d$ with $a, b, c, d \geq 1$ and $a > d$.
- H generates every string $0^a 1^b 0^c 1^d$ with $a, b, c, d \geq 1$ and $a < d$.
- S generates every string $0^a 1^b 0^c 1^d$ with $a, b, c, d \geq 1$ such that $b = c$ whenever $a = d$, which is the language asked for.

Why this is correct. The condition “if $a = d$ then $b = c$ ” is the same as “ $a \neq d$ or $b = c$ ”, so the language is the union of three sets: the strings with $b = c$, the strings with $a > d$, and the strings with $a < d$. The three alternatives for S generate exactly those three sets.

- OMI generates $0^a 1^n 0^n 1^d$ for all $a, n, d \geq 1$, which is exactly the set of strings of the required shape with $b = c$.
- A derivation from G applies $G \rightarrow 0G1$ some $k \geq 0$ times and then $G \rightarrow 0OIO1$, producing

$$0^k (0 0^p 1^q 0^r 1) 1^k = 0^{k+1+p} 1^q 0^r 1^{k+1}$$

for some $p, q, r \geq 1$. So $a = k + 1 + p$, $b = q$, $c = r$, $d = k + 1$, and $a - d = p \geq 1$. Conversely, any string $0^a 1^b 0^c 1^d$ with $b, c \geq 1$ and $a > d \geq 1$ arises this way, by taking $k = d - 1$, $p = a - d$, $q = b$, and $r = c$.

- A derivation from H likewise produces

$$0^k (0 1^p 0^q 1^r 1) 1^k = 0^{k+1} 1^p 0^q 1^{r+1+k}$$

for some $p, q, r \geq 1$. So $a = k + 1$, $b = p$, $c = q$, $d = k + 1 + r$, and $d - a = r \geq 1$. Conversely, any string with $b, c \geq 1$ and $d > a \geq 1$ arises by taking $k = a - 1$, $p = b$, $q = c$, and $r = d - a$.

Every string generated by S therefore has the shape $0^a 1^b 0^c 1^d$ with all four exponents at least 1, and satisfies $a \neq d$ or $b = c$, so it is in the language. Conversely, a string of the language with $a = d$ must

have $b = c$ and is generated by OMI , and a string with $a \neq d$ is generated by G or by H . The grammar is ambiguous, since strings with $b = c$ and $a \neq d$ have derivations of both kinds, but that is permitted here.

Sources and collaborators: None.

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